When Do the Images Prove a Dent? Identifiability and Abstention in CAD-Referenced Blade Damage Estimation

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Abstract

Deciding from a few photographs whether an aero-engine blade is physically damaged is ill-posed: a local intensity change may be a dent, a stain, a lighting change or a camera error, and a flexible model can explain any of them as another. Anomaly detectors and CAD-conditioned reconstruction output anomaly maps or deformed surfaces, but none states whether the images support a physical change. We formulate CAD-referenced damage estimation as a calibrated hypothesis test that may abstain. Given a healthy CAD mesh and 2–7 calibrated views, our method fits a nuisance-only explanation (pose, light gains, a brightness field and a stain patch) through a differentiable renderer, proposes dents with a Rao score test, fits a parametric 3D dent, and reports it only when its evidence exceeds a split-conformal threshold calibrated on healthy and stained blades. On a sealed test of 30 unseen Rotor37 blades, it locates 51% of visible dents at a matched 5% false-alarm rate, against at most 19% for CAD-prior deformation fields and a lifted 2D detector, and finds 12 of 13 dents deeper than 0.2 mm. Textbook χ^2 and a-contrario thresholds yield 32% and 96% false alarms on the same fits. On six held-out industrial designs, two corrections reduce false alarms from 51% to 4%, and class-specific thresholds raise damage-type accuracy over dents, nicks and cracks to 81%. Depth, however, is not identifiable from two passive views: found dents are measured at a median 60% of their true depth. The results delineate which conclusions passive photographs can support.

1. Introduction

Compressor and turbine blades are inspected visually, and the inspector's question is physical: is the blade dented, where, and how deep? The damage of interest is small. On a blade imaged at 512×512 pixels, a 0.07 mm dent changes a few hundred pixels by two to seven grey levels, whereas a 1° camera error, a 10% lighting change or a surface stain changes thousands. An automated system must separate geometric from appearance change and must not report damage the images do not support.

The difficulty is identifiability. Image appearance is produced jointly by geometry, illumination, reflectance and camera, so a local change admits several explanations: if lighting and appearance may vary freely they explain any dent away, and if shape may vary freely it absorbs any stain. Current methods do not address this. Anomaly detectors, including pose- and illumination-agnostic ones [4, 14, 25], output pixel scores, and invariance to illumination removes part of the very signal a shallow dent produces. CAD-conditioned reconstruction [20, 24] fits deformation fields that absorb whatever mismatch remains. In our own experiments, a render-and-compare fit produced false dents on 14 of 14 healthy or stained blades under a 1° camera error, and in 8 of 12 depth sweeps the best image match selected the wrong 3D shape: better image agreement is not better geometry.

Our key idea is to treat damage estimation as a calibrated hypothesis test that is allowed to abstain (Fig. 1). We make the acquisition constraints explicit — a calibrated rig with small pose corrections, fixed lights with free gains, and appearance changes only through a smooth brightness field and a local albedo patch — and compare a nuisance-only explanation H_0 with the same explanation plus a 3D dent, H_1. A dent is reported, with its location and depth, only if its evidence $L(H_0) - L(H_1)$ exceeds a threshold calibrated on real negatives; otherwise the method states that the views do not support a physical change.

Three design choices make the test discriminative. First, nuisance and damage are separated by the evidence they use: pose is fitted from silhouettes, which do not depend on lighting, and the dent is evaluated only on interior, non-grazing surface pixels. Sec-

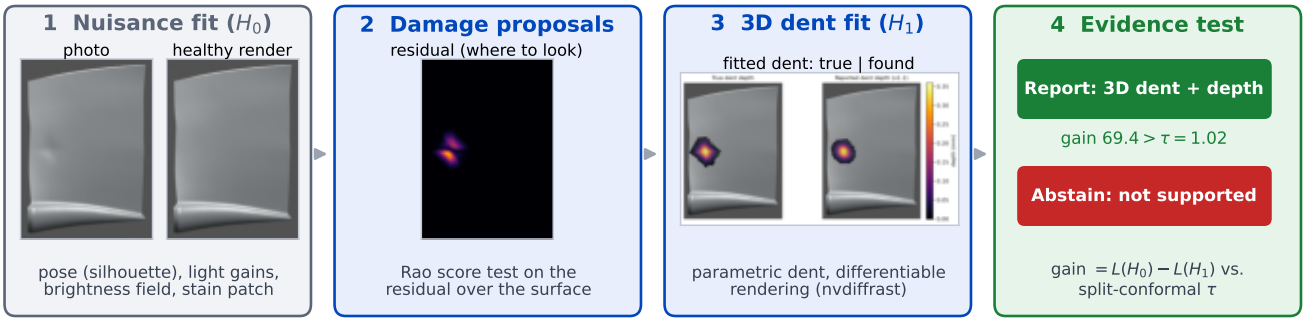


Figure 1. Overview. (1) The healthy CAD model is rendered and a nuisance-only explanation H_0 (camera pose from silhouettes, light gains, a smooth brightness field and a stain patch) is fitted to the photographs. (2) A Rao score test on the remaining residual proposes where a dent could be. (3) A parametric 3D dent is fitted there by differentiable rendering (H_1). (4) The dent is reported, with location and depth, only if its evidence $L(H_0) - L(H_1)$ exceeds a threshold calibrated on healthy and stained blades; otherwise the method abstains. Example from the sealed test: true depth 0.36 mm, measured 0.35 mm.

ond, a Rao score test evaluated over the whole surface with one backward pass through the renderer proposes dent locations, and dent and stain explanations are searched symmetrically so that neither can hide the other. Third, a light-linear neural shader makes rendered radiance exactly linear in the light gains, which prevents lighting changes from appearing as shape. The same fit predicts which dents the views can reveal, and physical specialists with class-specific thresholds assign a damage type.

Real photographs of damaged blades with measured 3D ground truth are not publicly available, so we build BladeDelta: 696 rendered cases on public Rotor37 compressor geometry [5] with exact damaged meshes, stain-only negatives, controlled camera and lighting perturbations and two sealed test splits, plus 720 cases on six industrial blade designs for transfer, defect types and counting. At a matched 5% false-alarm rate our method locates 51% of visible dents on the first sealed test, against at most 19% for re-implemented CAD-prior deformation fields and a lifted 2D detector, and a second sealed test reproduces the result. On held-out industrial designs, two corrections reduce false alarms from 51% to 4%; found dents, however, are measured at a median 60% of their true depth, and neither more views nor depth intervals repair this.

In summary, we make four contributions:

- A formulation of CAD-referenced damage estimation as identifiability with abstention under explicit acquisition constraints, with metrics that separate localisation, shape accuracy, false physical change and abstention on unobservable damage.
- An estimator combining silhouette-only pose, interior-domain nested testing through a differentiable renderer, score-test proposals and split-conformal calibration, which locates 51% of visible dents at 5% false alarms (baselines ≤19%), together with a profiled-information predictor of detectability (AUROC 0.90–0.95).
- Evidence that calibration is necessary rather than cosmetic: textbook χ^2 and a-contrario thresholds give 32% and 96% false alarms on the same fits, and class-specific thresholds lower false alarms on held-out designs from 10% to 4% while raising damage-type accuracy from 73% to 81%.
- BladeDelta, a benchmark of 1,416 rendered cases on public and industrial blade geometry, and an identifiability analysis showing that presence and location transfer across designs while depth and the type of irregular damage do not.

2. Related Work

Industrial and 3D anomaly detection. Memory-bank methods such as PatchCore [21] localise anomalous pixels; MVTec 3D-AD [2] and Anomaly-ShapeNet [16] add measured or synthetic 3D. Pose-agnostic benchmarks [27] and Gaussian-splat references [14] relax the fixed viewpoint, PIAD [25] adds illumination invariance, and M^{2}AD [4] benchmarks view–illumination interplay. Multi-view settings include RAD [6] and SiM3D [8], which lifts per-view scores into an anomaly volume; adding views can leak information in multi-view evaluation [7], and synthetic anomalies have been argued sufficient for 3D detection [22, 23]. Unlike these methods, which output anomaly scores and gain robustness by discarding illumination, we model illumination explicitly, estimate a physical surface change in millimetres, and state when the views do not support one.

Reference shapes and blade inspection. Looking 3D [3] aligns a query photograph with a reference shape and localises abnormal regions with boxes, and model-free detection reconstructs a normal shape by GAN inversion [18]. BladeSynth [10] renders procedural nicks, dents, scratches and corrosion on aero-engine blades to train 2D detectors. Unlike these, we use the exact design of the inspected part and output a 3D displacement field with a calibrated decision; we compare against a 2D detector trained on in-domain renders (Sec. 5).

CAD-conditioned reconstruction. CAD-NeRF [24] retrieves a CAD model and jointly optimises deformation, pose and appearance from few uncalibrated views; CADSplat [20] fits a shared deformation field to sparse posed views with a CAD prior and leaves measurement-quality deformed surfaces to future work. Unlike these, which treat deformation as a free component of a digital twin, we treat it as a hypothesis that must beat every nuisance explanation; our free-form and MLP deformation baselines re-implement this family.

Inverse rendering and uncertainty. Inverse rendering separates shape and material from images [19, 26], and radiance-field uncertainty [11, 13] and observation completeness [17] quantify where a reconstruction is unreliable. We build on differentiable rasterisation [15] with a learned shader on a known geometry; unlike uncertainty maps, our output is a test of a physical-change hypothesis against nuisance explanations.

Selective prediction and calibration. Selective classification abstains to guarantee a target risk on accepted inputs [12], and split-conformal quantiles provide distribution-free guarantees under exchangeability [1]; a-contrario detection controls false detections under an analytic noise model [9]. We calibrate the abstention threshold with split-conformal quantiles on real negatives; unlike classification, the accepted quantity is a physical surface change, and the score is the evidence of one hypothesis over a profiled nuisance explanation.

3. Method

Given a healthy mesh $\mathcal{M} = (V, F)$ with N vertices and K images $I_1, \dots, I_K$ with nominal intrinsics, extrinsics and lighting, we seek a displacement field $D \in \mathbb{R}^{N\times 3}$ on the airfoil together with a decision $s \in \{\text{supported}, \text{not supported}\}$. Our method (Fig. 1) renders the healthy mesh with a differentiable renderer, fits a nuisance-only explanation H_0, proposes dent locations with a score test, fits a dent explanation H_1, and accepts the dent only if its evidence exceeds a calibrated threshold. We describe the renderer and hypotheses, then proposals and the decision, and finally detectability and damage type.

Differentiable renderer. We rasterise $\mathcal{M}+D$ with nvdiffrast [15] using smooth vertex normals. A neural shader maps the per-pixel world normal n, position x and view direction ω to one radiance component per light, $R_k(n, x, \omega)$, and an environment term R_0. Because light transport is linear in each light's intensity, the pixel colour is $T\big(a \odot (R_0 + \sum_k g_k R_k)\big)$ for albedo multiplier a, light gains g and a learned monotone tone curve T. The shader is trained once on renders of geometries disjoint from all evaluation geometries, with a paired loss on image differences between lighting settings. The healthy CAD rendered at the nominal capture gives a per-pixel correction $B = I^{\text{ref}} - \hat{I}(\mathcal{M}; \eta_0)$ for static shader error. On a held-out blade the mean error is 0.92 grey levels, and image changes caused by lighting or by a dent correlate with the reference renderer at 0.90–0.98.

Nuisance hypothesis H_0. The nuisance η consists of a per-view pose correction (a rotation about the part and a camera-frame translation), three light gains, a low-frequency multiplicative brightness field per view and an optional albedo patch. Each view's pose is initialised by Euclidean ECC alignment with the reference image and refined by matching anti-aliased silhouettes to the observed foreground, coarse to fine; silhouettes do not depend on lighting, so the pose cannot compensate for a brightness change. All further comparisons use a fixed interior domain Ω: airfoil pixels at least 6 px from any silhouette or self-occlusion edge, excluding the blade root and grazing pixels ($|n \cdot \omega| < 0.35$), so both hypotheses are evaluated on the same pixels.

Dent hypothesis H_1. H_1 adds a dent with sub-vertex centre c (anchor vertex plus tangent offset), depth $\delta \geq 0$, radius r, aspect ratio, orientation, profile exponent p and signed rim coefficient κ:

$$D_i = -\delta \left[(1 - s_i^2)_+^p - \tfrac{\kappa}{2} \rho(s_i) \right] n_c, \qquad (1)$$

where s_i is the anisotropic normalised tangent-plane distance of vertex i from c, ρ a ring profile and n_c the centre normal, restricted to same-side airfoil vertices. None of the benchmark's truth families matches this profile exactly. Parameters are fitted with Adam through the renderer, jointly with the light gains and brightness field.

Score-test proposals. Let $h \in \mathbb{R}^N$ be an inward normal displacement; one backward pass gives $g = -\partial L_\Omega/\partial h$ at $h = 0$. For a kernel $w_{c,r}$ centred at c with radius r, $\langle w_{c,r}, g\rangle/\|w_{c,r}\|$ is the Rao score statistic for a small dent at (c, r) against H_0. We evaluate it for all centres and two radii with one sparse matrix product and start H_1 fits from the top-k after non-maximum suppression. Each start is fitted with and without the H_0 albedo patch, so a colour explanation fitted first cannot hide a dent.

Decision. With a Charbonnier loss ($\epsilon = 2/255$) summed over Ω, the evidence is

$$\Delta = L_\Omega(H_0) - L_\Omega(H_1) \geq 0, \tag{2}$$

since H_1 nests H_0. We report D if $\Delta > \tau$ and abstain otherwise, where τ is the split-conformal $(1-\alpha)$ quantile of Δ on calibration negatives (healthy and stain-only blades); this bounds the false-change rate on exchangeable negatives by α.

Predicting detectability. Linearising the rendered image at the H_0 fit in the depth h of a dent with unit shape u and in the nuisance parameters free during the test (light gains, brightness field and a stain at the same place), the evidence of a dent of depth h is approximately $\Delta(h) \approx \frac{1}{2}h^2 s$ with

$$s = j_d^\top W j_d - j_d^\top W J_\eta \big(J_\eta^\top W J_\eta + \Lambda\big)^{-1} J_\eta^\top W j_d, \tag{3}$$

where j_d and J_η are the image Jacobians, W the local curvature of the robust loss and Λ the nuisance prior. s is the dent's information after profiling out the nuisance; $h^* = \sqrt{2\tau/s}$ is the smallest depth that would be reported at that location.

Damage type and count. To name the damage, physical specialists are fitted at a reported location through the same renderer: the dent above, a nick (surface carved by a soft-edged ellipsoid at the edge), a crack (a thin dark line and a shallow groove along a fitted curved path) and a stain (albedo patch only). A judge selects the specialist with the lowest loss and outputs "type uncertain" when the runner-up is within τ. For class-specific thresholds, each specialist's evidence over H_0 is calibrated separately on negatives (τ_k); a defect is reported if any specialist exceeds its own τ_k, and the type is the one with the largest evidence. Defects are counted greedily: the strongest accepted dent is added to the model and the proposals are recomputed until none exceeds the threshold. The final output for a case is the vector (defective, type, intensity, count), with intensity the depth for dents and nicks and the length for cracks.

Split	Blades	Healthy	Stain	Dent	Dent + stain
Shader training	13	only for the renderer; never evaluated			
Development	12	12	24	36	24
Calibration	12	24	36	36	24
Test (sealed)	30	30	60	90	60
Test 2 (sealed)	30	30	60	90	60

Table 1. BladeDelta case counts. Each case has 7 rendered views; methods use 2 by default.

4. The BladeDelta Benchmark

Geometry. Healthy blades are public Rotor37 compressor geometries [5] (CC BY-SA 4.0), normalised to unit span and closed with a labelled artificial hub collar. Each mesh has 29,989 vertices. Shapes 0–12 were used during earlier development and only to train the shader. Shapes 13–24 form the development split, 25–36 the calibration split and 37–66 the sealed test split. Shapes 67–96 form a second sealed test split, rendered only after the post-hoc variants of Section 5 were frozen; 97–109 remain in reserve.

Damage. Each blade receives one of: no change, a colour-only patch (a soft darkened or rust-tinted albedo change), a dent, or a dent plus a colour patch (half overlapping). Dents follow one of three families — a cubic profile with a raised rim, a Gaussian, and an anisotropic elliptic profile (aspect 0.4–1) — with depth log-uniform in $[0.0005, 0.006]$ blade spans (about 0.04–0.45 mm on a 74 mm Rotor37 blade) and radius in $[0.04, 0.12]$. Centres face at least one of seven cameras and lie at least $\max(0.1, 1.2r)$ from the airfoil boundary.

Capture. Images are rendered with Blender Cycles (32 samples, denoised, AgX tone mapping, 512×512) from seven cameras on a ring around the blade, under one of four conditions: nominal; camera jitter (yaw, pitch and roll each uniform in $\pm 1.5°$ per view); light jitter (each of three area lights $\times[0.85, 1.15]$); or both. Methods receive only the nominal cameras and lighting. The default input is two views (60° and 120°); four- and seven-view settings are ablations.

Protocol. The split, case lists, metrics and hypotheses were hash-frozen before any test image was rendered; protocol changes are logged with dates. For each damaged case we record observability, the depth-weighted fraction of the dent visible in the input views.

CAD6 transfer split. To test transfer beyond Rotor37 we add six industrial blade designs supplied as STEP

solids (airfoil spans 24–70 mm), with platforms, fillets and dovetail roots. The airfoil is the largest B-rep face (two faces on one blade); everything else is excluded from evidence like the Rotor37 collar. Each blade is normalised to unit airfoil span in the Rotor37 frame and receives 76 cases (456 in total): 30% dents centred where both input cameras see the surface, 15% dents hidden from both, 20% healthy, 15% stain only, 10% dent plus stain and 10% healthy with camera error alone ($\pm 0.5°$, $\pm 1°$, $\pm 1.5°$). Dents are sized in millimetres as an inspector would specify them: depth log-uniform in $[0.1, 1.0]$ mm and radius in $[1, 5]$ mm. The case list was hashed before rendering, and no component — shader, threshold or hyperparameter — was refit on this split.

Defect types, counting and irregular damage. On held-out designs 3–6 we render 136 further cases (dents, stains and healthy blades, with simplified edge dents and grooves) and 64 cases with realistic defects: sharp nicks carved from the leading or trailing edge (0.5–2 mm deep) and hairline cracks drawn as thin dark lines with a 0.02–0.05 mm groove (5–15 mm long). For counting, 40 blades carry two or three separated defects; for irregular damage, 24 blades carry a dent creased by a sharp fold. All defects face both input cameras.

Metrics. Relative local error is the RMS vertex error over the true dent support divided by that of the unchanged CAD (1: no better than CAD; 0: exact). We also report relative whole-airfoil error, depth error, centre error, detection, false physical change on negatives (any vertex moved more than 2×10^{-5}) and abstention on unobservable dents, with 95% bootstrap intervals resampled by blade shape.

5. Experiments

Baselines. Unchanged CAD reports the healthy mesh and is the floor. Discrete search is our earlier render-and-compare method: image-guided proposals and a fixed profile at nominal pose, without nuisance fitting. Free-form field (per-vertex normal displacement with Laplacian smoothness and ℓ_1 sparsity) and MLP deformation field (a coordinate MLP with positional encoding) are the closest 3D CAD-based alternatives: they re-implement the core idea of CADSplat [20], a smooth deformation field fitted to sparse posed views under a CAD prior, within our renderer, since the official code was not released. 2D detector, lifted is a U-Net trained on (photo, healthy render) pairs of separate blades to segment dents, back-projected through the nominal cameras; it tests whether supervised 2D detection suffices. All methods share renderer, cameras and observations, and each thresholds its own score by the same calibration. For defect types we add YOLOv8n as a standard modern 2D detector, and for the decision rule textbook χ^2 and a-contrario thresholds.

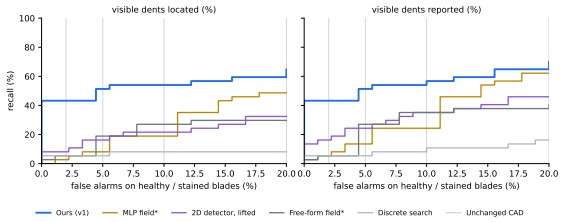


Figure 2. Recall of visible dents against the false-alarm rate on healthy and stain-only blades as each method's threshold is swept on the sealed test (left: located within 0.1 span; right: reported anywhere). *Re-implementations of the core idea of CADSplat [20].

Main results. As shown in Table 2, our method separates visible dents from negatives best (AUROC 0.82) and locates 54% of visible dents at its calibrated threshold, while every baseline locates at most 16%. The evidence test is what makes these claims trustworthy: without it, the same fits raise false alarms from 10% to 32% and wrongly claim 55% of hidden dents instead of 15%. Reported dents are measured to a median depth error of 0.03 mm and a median centre error of 0.13 mm (mean 0.77 mm; vertex spacing 0.73 mm). Recall is governed by depth (Table 3): 12 of 13 visible dents deeper than 0.2 mm are found, all at the correct location, whereas most dents below 0.1 mm, which change the image by about one grey level, are declined. A second sealed test on 30 further blades reproduces the result (Table 4; AUROC 0.79, 46% located, 6% false alarms).

Comparison at matched false-alarm rates. Because each method in Table 2 uses its own calibrated threshold, Fig. 2 compares them at equal false-alarm rates. At every rate from 2% to 20% our method locates more visible dents than any baseline (43% at 2%, 51% at 5%), while the strongest baselines locate 8–19% at 5%. The deformation fields catch up only at high false-alarm rates, because they explain stains as shape and must be thresholded above them; stains cause 8 of our 9 false alarms as well.

Calibration is necessary. The evidence Δ is a likelihood ratio, so a textbook threshold might seem sufficient. Table 5 shows it is not: a χ^2_8 threshold de-

Method	AUROC ↑	Found % ↑	Located % ↑	False alarm % ↓	Abstain (hidden) % ↑	Depth err. (mm) ↓	Centre err. (mm) ↓
Ours (v1, frozen)	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]	0.03	0.13
without the evidence test	0.82 [0.71, 0.90]	84 [68, 95]	70 [56, 84]	32 [21, 43]	45 [36, 54]	0.03	0.68
2D detector, lifted	0.71 [0.63, 0.80]	24 [12, 40]	16 [5, 30]	4 [1, 9]	96 [93, 99]	–	1.82
MLP deformation field†	0.74 [0.65, 0.81]	16 [6, 29]	11 [3, 22]	6 [1, 10]	100 [100, 100]	–	1.58
Free-form field†	0.63 [0.54, 0.70]	14 [4, 24]	11 [3, 20]	4 [1, 9]	99 [97, 100]	–	4.69
Discrete search	0.53 [0.43, 0.65]	5 [0, 13]	5 [0, 13]	6 [1, 10]	98 [95, 100]	–	1.60
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]	–	–

Table 2. Sealed test (240 cases, 30 unseen Rotor37 blades). Found / located: visible dents (≥50% visible in the two input views) reported / reported within 0.1 span of the true centre. Each method uses its own threshold, calibrated on the calibration split (split-conformal, $\alpha = 0.05$); "without the evidence test" reports every fit. Errors are medians over reported dents. Mean [95% interval, resampling blades]. †Reimplementations of the core idea of CADSplat [20], not the official model.

True depth (mm)	2 views	2 views (same cases)	4 views	7 views
< 0.1	1/10 (1 loc.)	0/4 (0 loc.)	1/5 (0 loc.)	3/9 (3 loc.)
0.1–0.2	7/14 (7 loc.)	1/2 (1 loc.)	2/6 (2 loc.)	7/11 (7 loc.)
> 0.2	12/13 (12 loc.)	4/4 (4 loc.)	4/6 (4 loc.)	8/9 (6 loc.)
All visible dents	20/37 (20 loc.)	5/10 (5 loc.)	7/17 (6 loc.)	18/29 (16 loc.)

Table 3. Visible test dents reported (located within 0.1 span) by true depth, at the frozen threshold. Below 0.1 mm a dent changes the image by about one grey level; the method declines rather than guess.

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
v1 (frozen)	0.79 [0.68, 0.88]	54 [40, 69]	46 [32, 60]	6 [1, 11]	88 [81, 93]
v1.1 (camera-check + correction fix)	0.80 [0.70, 0.88]	52 [38, 67]	46 [32, 61]	8 [2, 14]	88 [81, 93]

Table 4. Second sealed test: 30 further Rotor37 blades (67–96), 240 cases, unused by any earlier experiment. Each method ran once with its frozen threshold.

Decision rule	Target FA (%)	Actual FA (%)	Visible dents found (%)
Ours: conformal on 60 calibration negatives	5	10	54
Ours: conformal on 360 calibration negatives	2	4	46
Textbook χ^2_8 likelihood-ratio threshold	5	32	70
Textbook χ^2_8 likelihood-ratio threshold	1	32	70
A-contrario detector on the H_0 residual (NFA < 1)	≈0	96	100

Table 5. Calibrated versus textbook decision rules on the same sealed-test fits. FA: claims on the 90 healthy or stain-only cases. Found: visible dents located (for the a-contrario detector, any detection). The textbook rules assume i.i.d. image noise; the real nuisance mismatch is far larger, so they cannot control false alarms.

rived from the per-view residual noise and set for 1% false alarms flags 32% of healthy and stain-only blades, and an a-contrario detector on the H_0 residual [9] flags 96%. Both assume i.i.d. pixel noise, whereas the residual after the best nuisance fit is dominated by structured mismatch: residual pose error, lighting, and the gap between the learned shader and the renderer. The split-conformal threshold is set on that mismatch and stays near its target (10% at a nominal 5% with 60 calibration negatives; 4% at 2% with 360).

Ablations. Table 6 removes one component at a time on the development subset. Score-test proposals are essential: without them the fits start in the wrong place and AUROC falls from 0.855 to 0.420. Restricting evidence to the interior domain (0.685 on the whole image) and correcting the camera pose (0.679 without) are the next most important, since silhouette pixels and misregistration otherwise produce evidence for spurious dents. Removing the stain model raises the located rate (83% vs. 72%) but lowers AUROC to 0.753, because stains are then explained as dents and false alarms rise; the low-frequency brightness field has no measurable effect on this subset (0.846). More views make more dents observable (Table 7: 29 of 60 with seven views vs. 10 with two), but separation does not increase monotonically. A pre-registered attempt to cut false alarms with more appearance freedom or a dent-size prior failed (supplementary).

Variant	AUROC ↑	Located % ↑
Full method	0.855	72
No camera (pose) correction	0.679	72
No stain (colour) model	0.753	83
Whole image instead of interior evidence	0.685	61
No score-test proposals	0.420	6
No low-frequency brightness field	0.846	72

Table 6. Ablations on the development subset (18 observable dents, 18 healthy or stained blades).

Views	Observable dents	AUROC ↑	Located % ↑	Median rel. local err. ↓
2 views	10/60	0.847	60	0.43
4 views	17/60	0.777	41	1.00
7 views	29/60	0.857	62	0.47

Table 7. Number of input views (development subset). More views make more dents observable; separation and localisation are not monotonic in the number of views.

Predicting detectability. The profiled information s separates dents the method finds from those it misses with AUROC 0.90 on the sealed test and 0.95 on the second, against 0.85 for the best depth-and-visibility heuristic. Aggregated into per-point maps of the min-

Method	AUROC	Found (%)	Located (%)	False alarm (%)	Abstain hidden (%)
Ours, Rotor37 sealed test	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	10 [4, 17]	85 [77, 91]
Ours	0.74 [0.67, 0.84]	94 [89, 98]	70 [60, 81]	67 [36, 97]	33 [3, 67]
2D detector, lifted	0.86 [0.83, 0.89]	42 [35, 49]	36 [30, 42]	6 [3, 8]	100 [100, 100]
MLP deformation field	0.73 [0.68, 0.81]	84 [70, 97]	54 [47, 62]	43 [27, 58]	67 [48, 85]
Free-form field	0.60 [0.56, 0.66]	61 [36, 84]	37 [21, 53]	39 [20, 58]	64 [48, 82]
Discrete search	0.65 [0.58, 0.72]	16 [12, 18]	13 [10, 17]	4 [0, 10]	97 [94, 100]
Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]	100 [100, 100]
Ours v1 (frozen), held-out blades 3–6	0.72 [0.65, 0.90]	90 [85, 95]	67 [55, 79]	51 [10, 91]	50 [9, 91]
Ours v1.1, held-out blades 3–6	0.95 [0.93, 0.98]	86 [77, 95]	83 [76, 90]	4 [0, 11]	98 [93, 100]
Ours, CAD6 leave-one-blade-out τ	0.70 [0.63, 0.84]	13 [3, 26]	6 [1, 13]	4 [0, 11]	95 [89, 100]

Table 8. Transfer to six unseen real blade designs (CAD6, 456 cases; dents 0.1–1.0 mm deep). Every method is frozen: shader trained on Rotor37 only, thresholds from the Rotor37 calibration split. Top row: ours on the Rotor37 sealed test, for the transfer gap. Brackets: 95% intervals resampling blades.

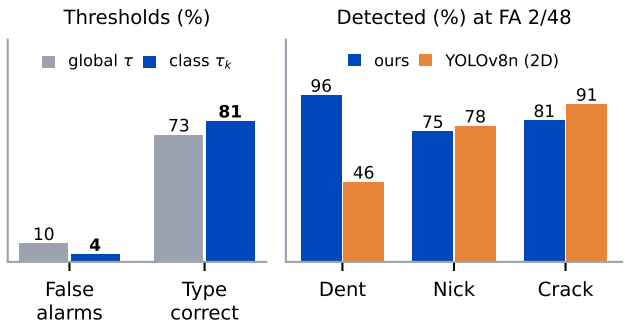


Figure 3. Held-out industrial designs. Left: a separate threshold per specialist (τ_k, calibrated on development blades) lowers false alarms on 48 healthy and stained blades and raises damage-type accuracy. Right: defects detected at the same false-alarm rate (2/48) by our method and by YOLOv8n trained on in-domain renders (mAP@50 0.75, ≈2 ms per image), which outputs 2D boxes only.

imum detectable depth h^*, the same quantity is optimistic for shallow dents (it implies 49–66% recall below 0.1 mm where 10% is achieved), because it assumes the search reaches the right location; we therefore treat the maps as an upper bound.

Transfer to unseen blade designs. Table 8 applies every method, unchanged, to the six industrial designs (456 cases). Separation falls from 0.82 to 0.74, and with the threshold calibrated on Rotor37 the frozen method raises false alarms on 67% of negatives, more on smaller blades (Spearman $\rho = -0.88$ with airfoil span). We traced this to two causes: the silhouette pose fit moves a correct camera by about 0.3° on geometry whose reference differs from the shader, and the image-space reference correction stays fixed while the blade moves under camera error. v1.1 applies the reference correction to the silhouette as well as to shading and reprojects the shading correction through the surface; both changes were developed on blades 1–2 only and frozen before blades 3–6 were run. On those held-out blades v1.1 raises AUROC from 0.72 to 0.95, cuts false alarms from 51% to 4%, locates 83% of visible dents (v1: 67%) and abstains on 98% of hidden dents (v1: 50%), without retraining and with comparable results on Rotor37 (Table 4).

Damage types and a 2D detector. On held-out designs the frozen v1.1 detector locates 23/24 dents, 21/32 sharp nicks and 22/32 hairline cracks without being designed for nicks or cracks. The specialists then label 22/22 located cracks and 9/15 located nicks correctly, with 6 further nicks marked "type uncertain", and preserve the location in 42 of 43 cases. As shown in Fig. 3, class-specific thresholds reduce false alarms from 5/48 to 2/48 at unchanged detection and raise type accuracy from 73% to 81%. At the same false-alarm rate, YOLOv8n trained on renders of the development blades is three orders of magnitude faster and detects as many nicks (25 vs. 24 of 32) and slightly more cracks (29 vs. 26), but far fewer dents (11 vs. 23 of 24), which change shading without sharp edges; it provides neither 3D location, depth nor abstention, and it is evaluated on images from its own renderer. Greedy counting gives the exact number of defects on 67% of 64 blades (healthy 8/8 and stained 7/8 correctly zero) and under-counts edge nicks on blades with three defects.

What is identifiable: presence versus depth. Presence and location transfer across designs; depth does not. Found dents on held-out designs are measured at a median 60% of their true depth (89% on Rotor37). Four views instead of two find more dents (45 vs. 36 of 60) but do not improve depth (median ratio 0.68 vs. 0.71 on dents found by both), and profile-likelihood depth intervals, which would be wide if depth were merely ambiguous, are narrow and contain the true depth in only 36 of 109 cases: the error is a systematic bias on unseen designs, not a search failure. Type abstention has a similar limit: on 24 folded dents, all located, the judge never abstains (12 crack, 8 dent, 4 nick), and an absolute fit-quality rule does not separate them from regular defects (the winning specialist explains a median 48% of the H_0 mismatch on folded dents and 49–68% on regular ones).

Qualitative results and real objects. Fig. 4 shows typical outcomes, including a miss and a false alarm. No public dataset pairs real blade photographs with measured damage geometry, so we also test on SiM3D [8]: six wooden stools scanned before and after damage, with the healthy scan as the CAD model, the healthy

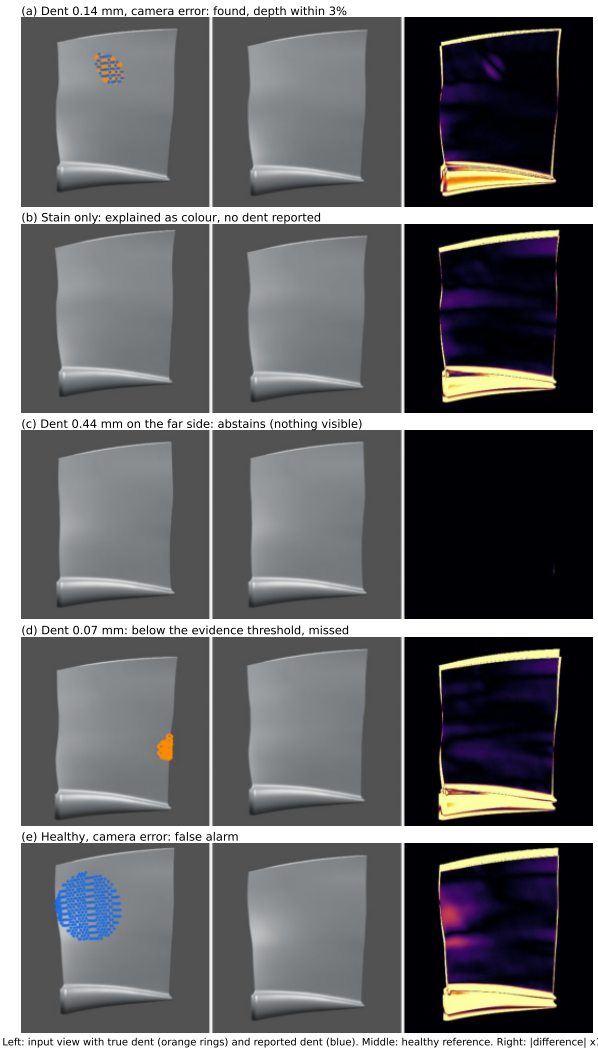


Figure 4. Sealed-test outcomes. (a) A 0.14 mm dent under camera error is found and measured. (b) A stain is explained as an albedo change. (c) A dent facing away from both cameras yields no evidence and the method abstains. (d) A 0.07 mm dent stays below the threshold. (e) A healthy blade under camera error is reported as dented, the residual failure mode.

Objects / defects / negatives	6 / 22 / 31
AUROC (defect vs. undamaged)	0.540
Detected (leave-one-object-out)	0/22
Located within 15 mm	0/22
False alarms	1/31

Table 9. Real objects (SiM3D) scanned before and after damage; truth from the measured scans.

photograph as the reference and the scan difference as the truth. On these textured objects the method does not separate defects from undamaged areas (Table 9; AUROC 0.54, 0 of 22 detected): the residual between sessions is dominated by wood-grain misregistration and specular changes that a shader trained on 12 photographs per object does not model. The evidence test is only as reliable as the appearance model behind it.

6. Limitations and Conclusion

Limitations. All positive results use rendered images; on real scanned objects (SiM3D) the method fails, and validation on real blades with measured damage remains to be done. Depth is biased low on unseen designs (median 60% of the true depth) and is not repaired by more views. The type judge does not abstain on shapes outside its specialist set (0 of 24 folded dents), counting under-counts edge nicks on blades with several defects, and a supervised 2D detector trained on in-domain renders matches our detection of nicks and cracks at a fraction of the cost. The conformal guarantee holds only for negatives exchangeable with the calibration set, so a new rig, lighting setup or blade family requires recalibration. The method answers whether these views support a physical change; it does not certify airworthiness.

Conclusion. We formulated CAD-referenced blade damage estimation as a question of identifiability under explicit acquisition constraints, answered by a calibrated test that reports a dent, with its location and depth, only when it explains the images better than every nuisance explanation, and abstains otherwise. On rendered blades this locates 51% of visible dents at 5% false alarms, where CAD-prior deformation fields and a lifted 2D detector locate at most 19%, and two corrections carry false-alarm control to unseen industrial designs (51% to 4%). Presence and location proved identifiable from two passive views, depth did not; controlled illumination and real printed, damaged and scanned blades are the next steps.

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